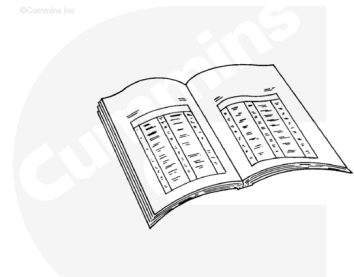


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

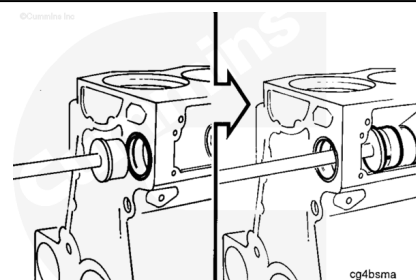
- Push the barring device shaft in and turn **counterclockwise** until the "I-6TC" mark on the damper is aligned with the pointer attached to the front of the block.
- Drain the oil from the lubricating oil pan. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Remove the cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Remove the flywheel housing and all related components. Refer to Procedure 001-020 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-020.html)
- Remove all of the rocker covers. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Remove all of the rocker assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Remove all of the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-

004-014.html)

- Remove all of the cam follower assemblies. Refer to Procedure 004-001 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-001.html)
- Remove the camshaft. Refer to Procedure 001-008 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-008.html)

Remove

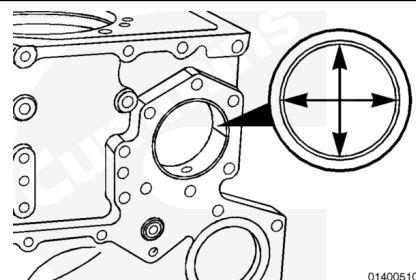
Remove the camshaft bushings using the camshaft bushing installation and removal kit, Part Number 3163015.



Inspect for Reuse

Measure the inside diameter of the camshaft bushings.

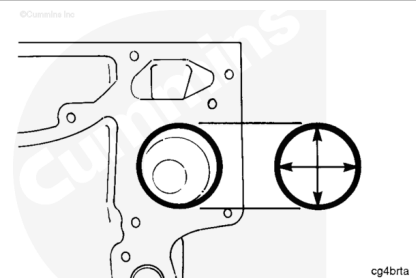
Camshaft Bushing Inside Diameter		
mm		in
105.01	MIN	4.134
105.10	MAX	4.138



If the bushing is **not** within specifications, the bushing **must** be replaced.

Clean and inspect the camshaft bores.

Camshaft Bushing Bore in the Block		
mm		in
111.966	MIN	4.408
111.988	MAX	4.409



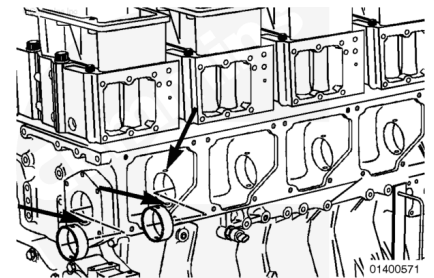
If the bore is damaged and can **not** be repaired, the cylinder block **must** be replaced.

Install

Verify oil rifle plugs are installed before installing camshaft bushings.

Align the oil holes in the bushing with the oil drilling in the block.

Position the notch in the bushing toward the front of the block.

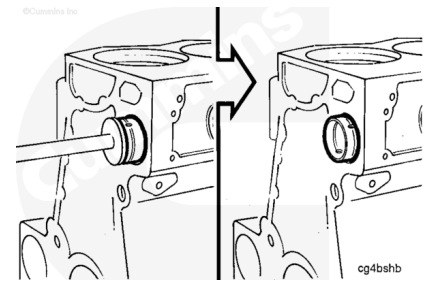


⚠ CAUTION ⚠

Clean the oil groove in the back of the camshaft bushing to prevent contamination of the lubrication system.

Install the bushing until the edge of the groove on the back (outside diameter) is even with the edge of the block.

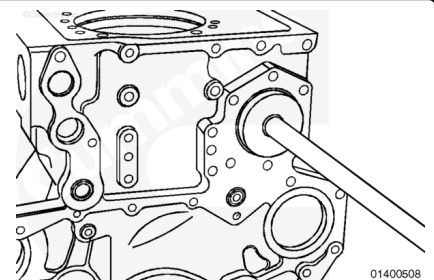
Clean the bushing outside diameter. Remove any metal shavings from the oil groove.



⚠ CAUTION ⚠

Be sure the front bushing is below the surface of the block. Engine damage can result.

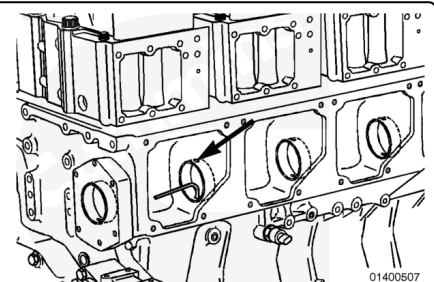
Install the bushing until the oil holes are aligned.



Use a 7-mm [$\frac{1}{4}$ -in] Allen wrench. Check the alignment of the bushing oil holes.

All holes **must** be aligned. The front bushing bore has **only** one drilling in the bottom of the bore, all other bores have two drillings, one in the top and one in the bottom.

There is no drilling to align with the upper hole on the front bushing.



Finishing Steps

- Install the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/89/89-001-008.html>).
- Install the camshaft cover plate. Refer to Procedure 001-011 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-011.html>)
- Install all of the cam follower assemblies. Refer to Procedure 004-001 in Section 4. (</qs3/pubsys2/xml/en/procedures/89/89-004-001.html>)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/89/89-004-014.html>)
- Install the rocker assemblies. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>)
- Adjust the overhead setting. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-006.html>)
- Install the rocker cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>)
- Install the cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)
- Install the flywheel housing and all related components. Refer to Procedure 001-020 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-020.html>)



